

Point of view: Suggestions for professionals looking to buy mid- or high-end boards

FPGA density (or how large of a design can fit in to the FPGA). This is more important for advanced-level boards.

Price and on-board peripherals. The less the number of peripherals already available on board, the more additional extension boards users may have to purchase, thereby resulting in higher total cost. For advanced-level boards, sometimes it may be desirable to not have a lot of on-board peripherals because the user can pick and choose the exact functionality by selecting or buying specific expansion boards. There is a cost versus flexibility case here.

Must-have features. Double data rate synchronous dynamic random access memory (DDR SDRAM) and a serial interface is a must if soft processor based (like, Microblaze) embedded system building is planned.

Ethernet. Especially important if planning to run Linux.

Configuration options. Built-in configuration system preferred since USB JTAG cables can be very expensive.

— Contributed by Anoop Jose, manager - research and development, Numato Systems Pvt Ltd

user group should consider mid- or high-end FPGA platforms with a broad category of features. It enables them to quickly prototype their sub-systems or system concept without being limited by the performance of FPGA platforms."

On the other hand, student communities use FPGA development platforms as a learning platform as well as a target platform for their projects and research activity. Kumar says, "This user group should consider easy-to-use, low-cost platform having additional features support through piggyback or daughter cards." He adds, "With this approach, a user can set up basic development with low cost and add additional interfaces as required, instead of investing into feature-rich, expensive hardware."

Later in the article we will take an in-depth look at parameters to be considered and some FPGA development boards available in the market that will help you make an informed decision before buying one. Before that, let us take a brief look at how today's FPGA boards are accelerating prototyping and designing.

Today's boards provide faster performance and time-to-market

Increasing FPGA densities, board interfaces, supported connectors for add-on modules from third-party vendors and built-in joint test action group (JTAG) interfaces are making

boards more powerful and capable. These are driving down the overall cost and providing improved timelines for product development. Natarajan M.M., vice president for South Asia of Arrow Asia-Pacific components business, says, "New-generation FPGA boards are equipped with high-speed mezzanine card (HSMC)/FPGA mezzanine card (FMC) connectors, allowing add-on kits from third-party boards to be plugged on base boards designed on Altera's FPGAs."

Kumar says, "Adding support for various configurations of external analogue-to-digital converter (ADC)/digital-to-analogue converter (DAC) cards, network processing modules and industrial control modules has enabled users to use a common platform with applications having varied signal acquisition, interfacing and performance targets." He adds, "With re-usability being a requirement today, supporting the use of a common FPGA based prototyping environment for different designs has significant cost and time-to-market advantages."

Integration of hard floating-point DSP blocks, 64-bit application class processors, up to 1GHz core fabric speed, high system interconnect bandwidth, C based modelling and hardware design with Altera software development kit (SDK) for Open Computing Language (OpenCL) allows for high-performance and power-efficient heterogeneous computing that meets

FPGA platform availability has improved very much in India today. In 2008, one had no option but to import low-cost FPGA boards

customers growing design challenges.

Bowers informs that their goal was to make Mercury development board as friendly as possible for engineers who want to include FPGAs in their projects, be it students making one-time projects or professionals working to prototype their designs. He says, "Users do not have to worry about soldering tiny FPGA packages, providing multiple voltage rails and numerous bypass capacitors, adding an ADC, soldering on a fast external memory or worrying about interfacing with 5V logic. Mercury takes care of all this, leaving the user to worry about the goals of the actual design."

Selecting the right board

Every user will have different needs based on the design, feels Bailey Hsu, director - marketing division, Terasic. Factors such as memory speed, capacity and compatibility with other hardware are pretty fundamental. He says, "The most important thing to look for is the kind of input/output (I/O) expansion on the FPGA development board. A good board must have advanced fundamental features that are accommodating for everyone and at the same time leave enough expansion space for users to connect and expand for their needs."

Let us take a look at the four most important factors to consider before buying a development board.

Features. The first and foremost factor that affects an engineer's development board selection is the feature set on offer. Bryan Fletcher, global technical marketing director, Avnet Electronics, says, "A board manufacturer helps engineers make this easier by offering multiple product offerings in a full portfolio of development boards and adoptable